

mooreVIEW Cloud — Vertical Shard Architecture

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Product: mooreVIEW Cloud SaaS
Audience: Platform operators, integrators, CFO / capacity planning
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Executive summary

mooreVIEW cloud scale-out uses **vertical ingest shards** (lift/septic/WWTP, pools, ALF, c-store/grocery) on top of a **fixed control plane** (SaaS + LB + shared Mongo + MQTT + archive). MQTT topics remain `mooreview/v1/{tenantId}/{deviceId}/...` — sharding is implemented via a **tenant** → **vertical** → **shard** registry, not a wire-protocol change.

Key rules:

- **LB on :3100 (SaaS)** — yes, from launch.
- **LB on :3090 (ingest)** — no for telemetry; optional tenant-aware proxy for Parc HTTP commands only.
- **One ingest cell ≈ 500K live cloud tags**; add cells within a vertical when a shard exceeds ~450K tags.
- **M18 hybrid:** 2 cells (field + retail). **M36 target:** ~11 cells across 4 verticals at 5M tags.

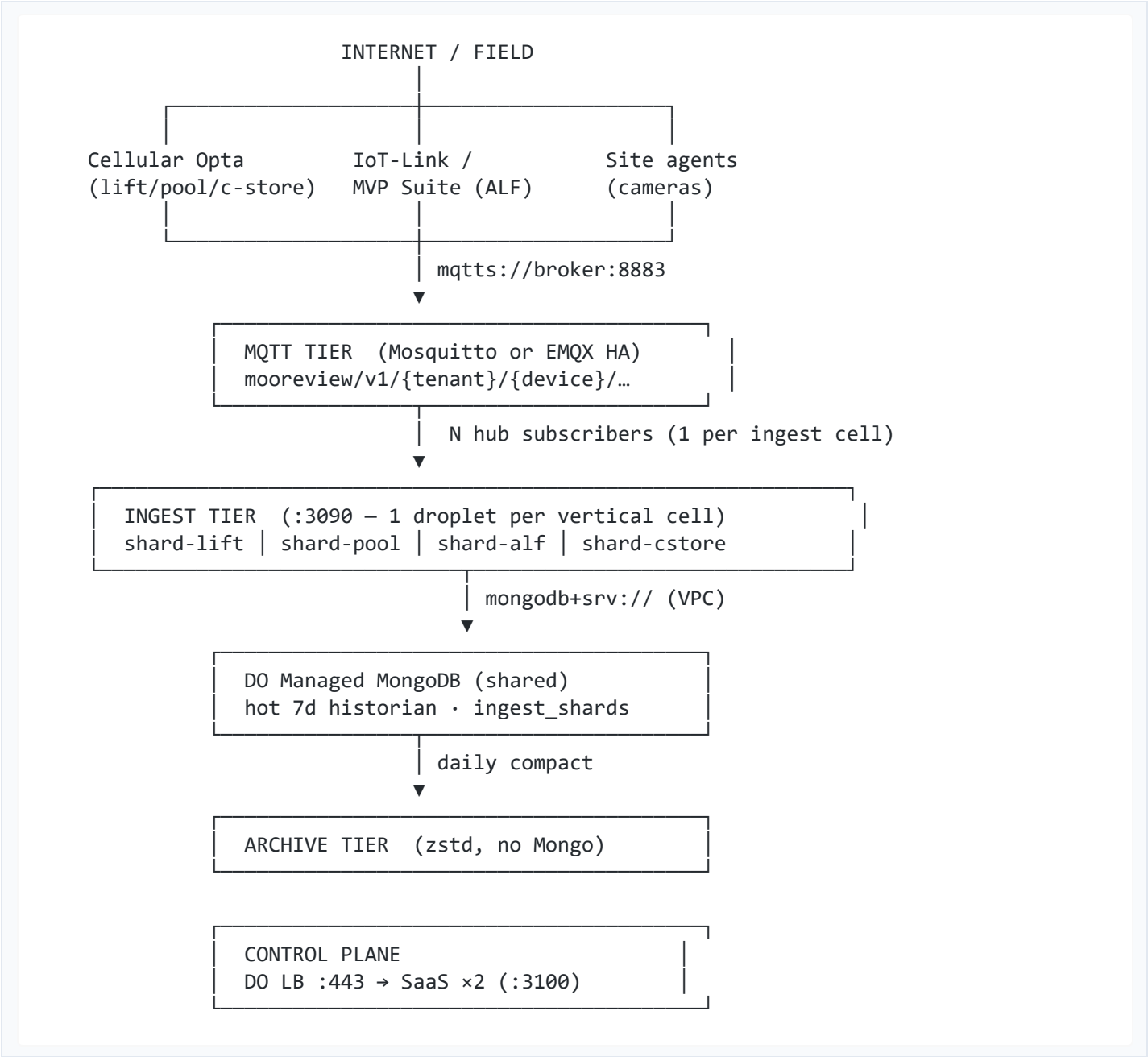
1. Reference fleet (planning baseline)

Assumptions: **5 min scheduled report + exception-on-change**; **ALF slim mirror** (400 cloud tags/campus, full model on appliance); pools **lift-like I/O** (85 tags/site).

Segment	M12	M18	M24	M36
Lift / septic / WWTP	2,000	3,000	13,100	30,100
Pools	667	1,000	4,380	10,000
C-store / grocery	333	500	2,568	6,600
ALF campus	33	50	256	660
Total sites	3,033	4,550	21,304	54,360
Cloud tags	277K	415K	1.94M	5.0M

Tag mix @ M18: lift **61%** · pool **21%** · c-store **12%** · ALF slim **5%**.

2. Full platform diagram



3. Vertical shards

Shard	Vertical	M18 tags	Reporting	Cloud profile
A	Lift / septic / WWTP	~255K	5 min + exception	Full mirror, PdM/ONNX
B	Pools	~85K	5 min + exception	Lift-like I/O
C	ALF campus	~20K	5 min + exception (slim)	Appliance-heavy, ~400 tags uplink
D	C-store / grocery	~50K	5 min + exception	~100 tags, cold chain

3.1 Shard A — Lift / septic / WWTP

Setting	Value
MOOREVIEW_INGEST_SHARD_ID	shard-lift
MOOREVIEW_VERTICAL	lift
MOOREVIEW_MAX_TAGS	525312
Droplet	32 GB / 8 vCPU
Host ONNX	on (host-supplement)
Historian pens	8/site @ 300 s

3.2 Shard B — Pools

Setting	Value
MOOREVIEW_INGEST_SHARD_ID	shard-pool
MOOREVIEW_VERTICAL	pool
MOOREVIEW_MAX_TAGS	131072
Droplet	16 GB
Historian pens	8/site (PH, ORP, flow, pump)

3.3 Shard C — ALF

Setting	Value
MOOREVIEW_INGEST_SHARD_ID	shard-alf
MOOREVIEW_VERTICAL	alf
MOOREVIEW_MAX_TAGS	26112
Droplet	8 GB
Host ONNX	off
Uplink	Appliance c1oudRemote slim ~400 tags

3.4 Shard D — C-store / grocery

Setting	Value
MOOREVIEW_INGEST_SHARD_ID	shard-cstore
MOOREVIEW_VERTICAL	cstore
MOOREVIEW_MAX_TAGS	65536

Droplet	16 GB
Historian pens	6/site (cooler, RTU, leak)

4. MQTT plumbing

Wire format (unchanged):

```
mooreview/v1/{tenantId}/{deviceId}/telemetry
mooreview/v1/{tenantId}/{deviceId}/online
```

Shard assignment via tenant metadata — each cell subscribes only to assigned tenants:

```
shard-lift:    mooreview/v1/ace-septic/#, mooreview/v1/putnam-county/#, ...
shard-pool:    mooreview/v1/pool-cloud/#, ...
shard-alf:     mooreview/v1/alf-sunrise/#, ...
shard-cstore:  mooreview/v1/circle-k-florida/#, ...
```

Do not run multiple ingest nodes with global `mooreview/v1/+telemetry` subscribe — that duplicates all processing.

5. HTTP / SaaS routing

```
POST /api/parc/devices/{deviceId}/cmd
  1. Session → tenantId
  2. tenant.vertical → ingest_shards.internalUrl
  3. Proxy → http://10.0.0.2X:3090/api/parc/devices/{id}/cmd

GET /fleet
  → parallel fan-out to all shards, OR denormalized fleet_devices in Mongo
```

Shard registry (Mongo `ingest_shards`):

```
{
  "shardId": "shard-lift",
  "vertical": "lift",
  "internalUrl": "http://10.0.0.21:3090",
  "tenants": ["ace-septic", "putnam-county-utilities"],
  "maxTags": 525312,
  "tagCount": 412000,
  "status": "active"
}
```

6. Mongo (shared)

Collection	Partition	Retention
tag_logs / tag_samples_ts	company, siteId, at	7 d hot → archive
edge_inference_ts	deviceId, at	7 d
ingest_shards	shardId	permanent
archive_registry	company, siteId	permanent

Fleet historian: **sampleIntervalMs: 300000** (match 5 min reports). Do not use 5 s default at fleet scale.

7. Scale-out by period

Period 1 — 0–12 months (ends ~277K tags)

Component	M0–M6	M6–M12
SaaS	1× 4 GB	LB + 2× 4 GB
Ingest	1× 16 GB (field)	1× 16 GB field + 1× 8 GB retail
MQTT	1× 4 GB	1× 4 GB
Mongo	entry tier	4 GB RAM / ~80 GB disk

Period 2 — 12–24 months (277K → 1.94M tags)

M18 (~415K): 2 cells — field 32 GB + retail 16 GB.

M24 (~1.94M): 6 cells:

Vertical	Tags	Cells
Lift / septic / WWTP	~1.19M	3× 32 GB
Pools	~398K	1× 16 GB

C-store	~233K	1 × 16 GB
ALF	~93K	1 × 8 GB

Period 3 — 24–36 months (1.94M → 5M tags)

Vertical	Cells @ 5M
Lift / septic / WWTP	6 × 32 GB
Pools	2 × 16 GB
C-store	2 × 16 GB
ALF	1 × 16 GB
Total ingest	11 droplets

MQTT: 2 × 16 GB (or EMQX HA) above ~50K connections.

8. Hybrid vs full vertical split

	M18 hybrid (recommended)	Full 4-vertical @ M18
Ingest droplets	2 (field + retail)	4
Extra cost	—	~+\$120–180/mo
Ops	Simpler	Best isolation

Recommendation: Start 2 shards @ M18; split pools to own shard when pool tags > 150K; full 4-vertical by M24.

9. Multi-vertical tenants

MQTT topics are tenant-scoped. Options:

Approach	When
Split tenant per vertical (acme-lift, acme-pool)	Recommended M18–M36
Route by device.category in SaaS	Single tenant, multiple verticals
Topic v2 mooreview/v2/{vertical}/...	Only if tenant-split outgrown

10. Software prerequisites

Item	Purpose
tenant.vertical	lift pool alf cstore
Mongo ingest_shards	Tenant → shard routing

MOOREVIEW_INGEST_TENANTS + filtered hub subscribe	No global +/telemetry
SaaS shard router	Parc cmds + live tags
Fleet aggregator	/fleet across shards
Slim ALF uplink on appliance	Keep ALF shard small

11. Private network (VPC)

Host	Role	Expose publicly?
LB + SaaS	Operators	443 only
MQTT	Devices	8883 (TLS)
Ingest shards	Internal	No
Mongo	Managed	No (allowlist)
Archive	Internal	No

12. Related documentation

Topic	Path
Cloud infra pricing (0–36 mo)	docs/marketing/CLOUD_INFRA_PRICING.md
DO SaaS deploy	docs/CLOUD_DEPLOY_DO.md
Archive pipeline	docs/ARCHIVE_EXPORT.md
AI / edge / cloud roles	docs/AI_EDGE_CLOUD.md
Infrastructure projections	docs/marketing/MooreVIEW-Infrastructure-Projections.md

mooreVIEW Cloud Vertical Shards v1.0 — platform planning.

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